

CS-4032 Web Programming

Assignment 03: Property Dealer CRM System

Roll Number: 23i-0829 | Section: B

Submitted To: Ms. Tajwar Mehmood

https://github.com/Zaid-fareed/WebProgramming/tree/main/i230829_b_a3

1. Project Overview

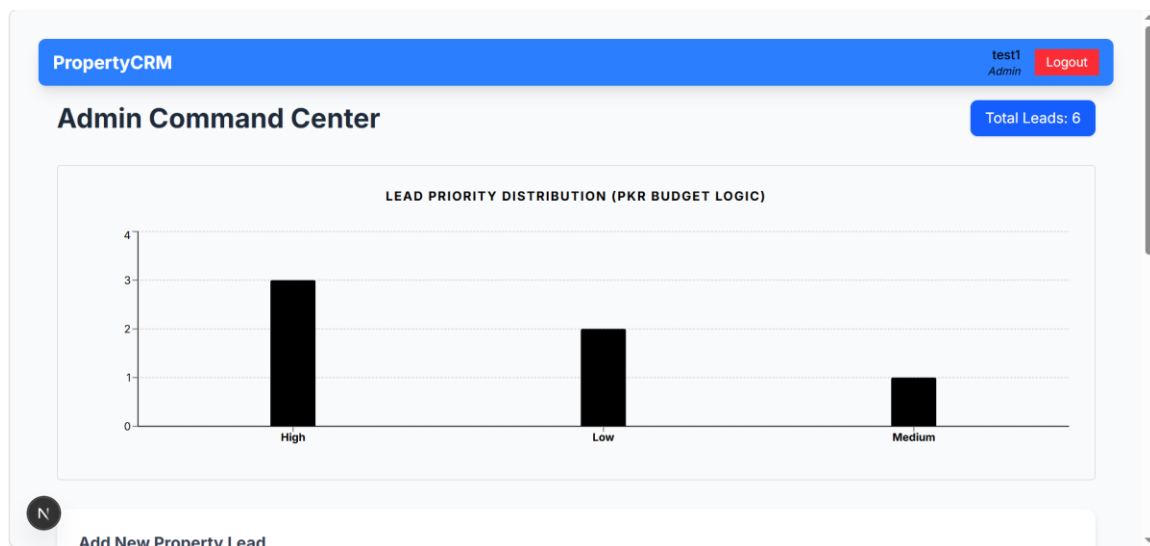
This project is a full-stack Property Dealer CRM built with Next.js, MongoDB, and Socket.io. It features role-based access control (RBAC), real-time notifications, automated email alerts, and a lead scoring system based on budget thresholds.

2. Implementation Details

- **Authentication & RBAC:** Implemented using NextAuth.js with JWT. Roles include Admin (full access) and Agent (assigned leads only).
- **Lead Scoring Logic:** Budget > 20M: High Priority; 10M-20M: Medium Priority; < 10M: Low Priority. Handled via backend logic.
- **Real-time Updates:** Socket.io integrated into a custom server.js for instant dashboard refreshes on assignment.
- **Security:** Rate limiting middleware restricts agents to 50 requests per minute to prevent API abuse.
- **External Integration:** WhatsApp click-to-chat and automated email notifications via Nodemailer/Ethereal.

3. System Screenshots

Admin Dashboard (Analytics & Charts)



Lead Management Table (Priority Highlighting)

Add New Property Lead

Client Name

Email Address

Phone (e.g. 923001234567)

Budget (PKR)

Apartment

Website

Create Lead & Calculate Priority

Lead Inventory

Name	Priority	Budget	Assign To	Actions
sdaewd	HIGH	215,115,816,186,161	agent2	WhatsApp History Delete
someone	MEDIUM	10,000,000	test	WhatsApp History Delete
someone	HIGH	100,000,000	agent2	WhatsApp History Delete
one	HIGH	200,000,000	test	WhatsApp History Delete
noone	LOW	12,321	agent2	WhatsApp History Delete
someone	LOW	5,000	test	WhatsApp History Delete

Agent Dashboard

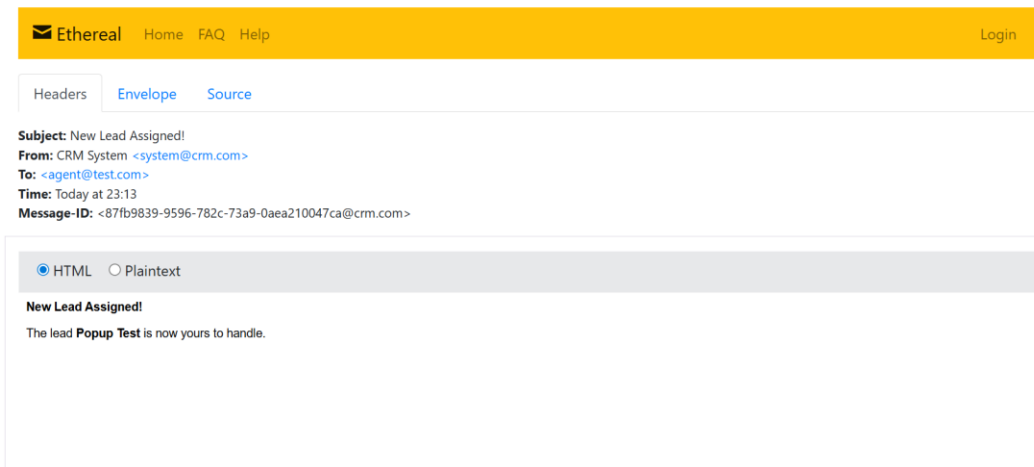
PropertyCRMagent2AgentLogout

MY ASSIGNED LEADS

Name	Priority	Budget	Actions
sdaewd	HIGH	215,115,816,186,161	WhatsApp History
someone	HIGH	100,000,000	WhatsApp History
noone	LOW	12,321	WhatsApp History

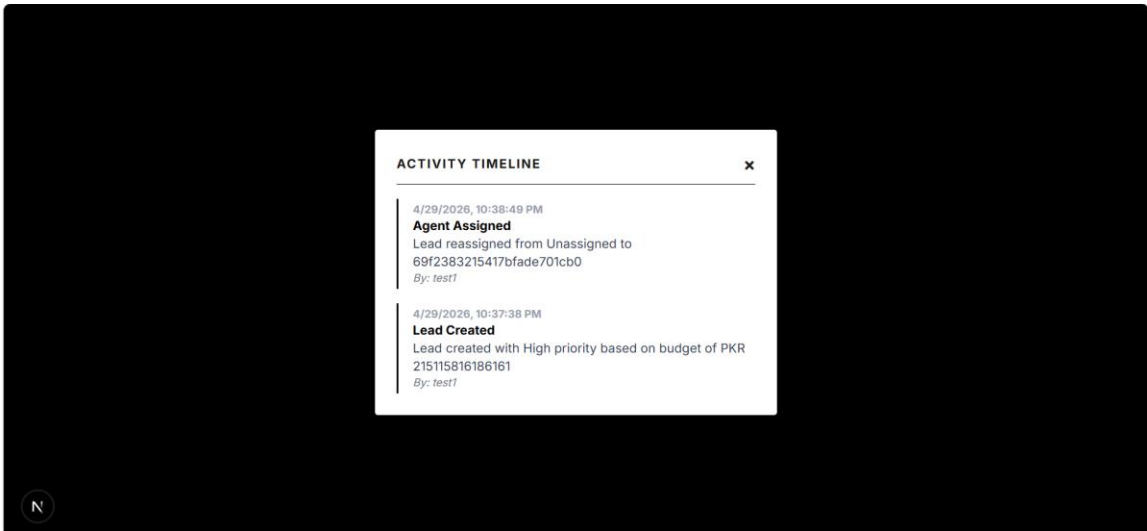
Email Notification Preview (Terminal/Ethereal Log)

When a lead is assigned to an agent, an email is sent to that agent



4. Bonus Features

- **Lead Activity Timeline:** A complete history tracking system storing every action in an ActivityLog model.



- **Bidirectional Real-time Updates:** Using Socket.io for live notifications across dashboards.

MY ASSIGNED LEADS

Name	Priority	Budget	Actions
Popup Test	HIGH	1,811,919,191	WhatsAppHistory
sdaewd	HIGH	215,115,816,186,161	WhatsAppHistory
someone	HIGH	100,000,000	WhatsAppHistory
noone	LOW	12,321	WhatsAppHistory